

Bosch Camera Stop Bar Settings REV-6

This Guide is intended to be used in conjunction with the specific cameras Installation and Operations manuals along with VCA Operation Manual and Configuration Manager Operations manual.

With the new firmware versions, you are now required to configure VCA (analytics) through Configuration Manager. I recommend making sure you are using the latest version of Configuration Manager, currently it's 7.78.0136 I also recommend using the latest camera firmware.

For best performance, I recommend that cameras are mounted at least twice as high as any vehicles. The new CPP14.2 cameras with the MVC-IVA-TRA traffic pack will detect vehicles or people whether stationary or otherwise.

By default, the cameras are set for DHCP. You will need to set a static IP for the camera(s). This can be done through Configuration Manager. Right click on the camera or cameras and select Device Network Settings and set the Network IP, Subnet and Gateway.

When first accessing a new camera, the first thing you must do, is enter a strong password (alpha-numeric and 12 characters or more) for the service user.

I recommend naming your cameras so you can differentiate them. You can do that in Configuration Manager under The General and Unit Access tabs or in the VMS. You will want to put the camera name in both the Video Name and Host Name fields.

General/Date/Time tab: Make sure the time and date are correct for your time zone. Add pool.ntp.org to the time server address or other valid NTP server.

General/Initialization tab: For Sensor Mode I always use 30fps HDR X unless 60fps is absolutely required. Depending on camera model, you will have a Lens Wizard or Camera View Wizard option. Use this to adjust your field of view and focus. Remember that the depth of field diminishes as the IRIS opens in low light situations. Set your focus window to just cover the detection zones to obtain optimum results. Also set Device Optimization Profile to IVA Optimized.

General/Positioning: Here is where you enter the height of the camera in meters over the field of view. This is not necessarily the height of the camera directly where it is located, due to varying terrain. After you enter the height of the camera make sure you refresh the sensor settings. You may also enter the GPS coordinates and compass heading of the field of view of the camera in decimal format.

General/Camera Calibration: for CPP14.2 series cameras and above with the Traffic Pack has a new feature. They now have Autocalibration and adds significant accuracy to the analytics. It simplifies install greatly.

Connectivity/Cloud Services: This will require access to the Internet. Set operation to ON save this, and select Register. Your Remote Portal login will be required to finish the Cloud connection.

Camera/Display: Under Stream Security set Video authentication to SHA-256 If recording.

Camera/Imaging: Set Current mode to Vibrant, this will be optimal for 90% of all applications. Under Enhance, set High dynamic range to HDR-X – Motion Optimized, Intelligent Defog to Auto. Under ALC, set Day-to-Night switchover to -10 and Night-to-Day switchover to -9.

Camera/Video Streams: This is controlled by the VMS and does not impact analytics. If you are recording to SD cards locally, select the stream that best fits your application. The higher the image quality the more storage will be utilized. If you are only recording locally for verification of IVA triggers, you can select Bit rate optimized and a frame rate fast enough to give you a couple of frames of video with the object in your detection zone.

Recording/Recording management: If you are using an SD card for video recording, select SD card for Preferred storage type, typically the SD card is configured under Secondary Recording (Primary is usually controlled by a VMS). Click on the play icon under the hard drive symbol on the right side to start recording.

Recording/Recording Profiles/Recording 2/Encoder Profile: Select the stream profile that best suits your application for the SD card.

Recording/Recording scheduler: I recommend using a single profile across all days and times.

VCA/Main Operation: Select Profile #1, and select IVA PRO Traffic for Analysis type. Under Tracking parameters, select Traffic tracking 3D.

VCA/Tasks: Select the Object in Field rule and name it the same as the detector zone in the MH Corbin box. Select the right arrow at the bottom. Create a new field that is very narrow over the stop bar detection zone. Placement will be determined by the view of the camera, but centered over the lane is most desired. Then create a second field that is the actual detection zone. I recommend that the 2 zones actually touch each other. I recommend setting debounce to .2 seconds and Intersection Trigger to Object Base Point. Select Next and select all fields in their specified order. Select the next arrow and Select all vehicles. Select the next arrow 2 times and make sure Motion type is set to All objects. Select next arrow twice to skip color filter. Under Define additional timer, set both to .1 seconds and select the check mark, the rule is complete.

VCA/Metadata: Deselect Polygon, Face Detection, Stationary Person.

Network/Network Access: Make sure the static IP is set, along with Subnet Mask, Gateway Address and DNS. I recommend setting Min. TLS version to 1.2+, everything else I leave at factory default.

Network/Network services: Select RTSP especially if using the MH Corbin SDLC interface.